PROJECTIVE AMPLITUDE OF A PSEUDO-COHERENT COMPLEX IS EQUIVALENT TO PERFECTION WITH BOUNDED TOR AMPLITUDE AND TO EXT VANISHING AGAINST FINITELY PRESENTED MODULES

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Editorial note. The author writes 'the assumed vanishing of ext groups' with a lowercase 'ext' in the counted proof span; that is the author's text and is reproduced unchanged. The six references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The characterisation of perfect complexes and the relation between projective amplitude and tor amplitude. invoked at proof lines 20153-20158. Establishes the equivalence of the first two conditions.
- Vanishing of Ext against finitely presented modules versus against arbitrary modules, for a pseudo-coherent K. used at proof lines 20161-20165. Is the reduction that brings conditions (3), (4) and (5) into the range of the projective-amplitude criterion.
- The projective-amplitude criterion in terms of Ext vanishing. invoked at proof lines 20166-20167. Gives the equivalence of (1) with the three numerical conditions.

Source reference. Original lemma 00HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 636–636. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 18889: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18889–18889. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 17414: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17414–17414. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16596: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16596–16596. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16906: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16906–16906. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let K be a pseudo-coherent object of $D(R)$. Let $a, b \in \mathbf{Z}$. The following are equivalent*

1. *K has projective-amplitude in $[a, b]$,*
2. *K is perfect of tor-amplitude in $[a, b]$,*
3. *$\mathrm{Ext}^i_R(K, N) = 0$ for all finitely presented R-modules N and all $i \notin [-b, -a]$,*
4. *$H^n(K) = 0$ for $n > b$ and $\mathrm{Ext}^i_R(K, N) = 0$ for all finitely presented R-modules N and all $i > -a$, and*
5. *$H^n(K) = 0$ for $n \notin [a - 1, b]$ and $\mathrm{Ext}^{-a+1}_R(K, N) = 0$ for all finitely presented R-modules N.*

Proof. From the final statement of Lemma 18889 we see that (2) implies (1). If (1) holds, then K can be represented by a complex of projective modules P^i with $P^i = 0$ for $i \notin [a, b]$. Since projective modules are flat (as summands of free modules), we see that K has tor-amplitude in $[a, b]$, see Lemma 16906. Thus by Lemma 18889 we see that (2) holds.

In conditions (3), (4), (5) the assumed vanishing of ext groups $\mathrm{Ext}^i_R(K, M)$ for M of finite presentation is equivalent to the vanishing for all R-modules M by Lemma 16596 and Algebra, Lemma 00HA. Thus the equivalence of (1), (3), (4), and (5) follows from Lemma 17414. □